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Process qualification of laser powder bed fusion based on processing-defect structure-fatigue properties in Ti-6Al-4V

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ABSTRACT

The aim of this manuscript is to give a compact overview of the results that illustrate the applicability of processing-structure-property relationships in the increasingly important context of 3D printing of metals. A process qualification approach based on the physics-based understanding of defect formation in laser powder bed fusion (L-PBF) additive manufacturing (AM) is investigated for an aerospace-grade titanium alloy (Ti-6Al-4V). A physically interpretable qualification approach is critical for enabling L-PBF part certification for structure-critical applications. This approach relies on systematic experimentation, characterization, testing, and data analysis tasks including design of experiments varying power and velocity to generate varying defect populations, process window development based on defect structure, high throughput fatigue testing, and fractography, 2D porosity characterization, and use of extreme value statistics to develop a porosity metric that, in turn, could have predictive power for the variation in fatigue performance. Results from four-point bend fatigue tests demonstrate that a process window can be defined based on this key mechanical property. This relatively high throughput approach can, in turn, support a reduced set of round bar fatigue tests typically used for qualification. Overall, the proposed ecosystem for process qualification of L-PBF AM shows promise and is expected to apply to other materials and powder bed fusion AM technologies.

1. Introduction

The recognition of the unique capabilities of laser powder bed fusion (L-PBF) additive manufacturing (AM) technologies has led to significant growth of research and development in the field over the last two decades. The growth is primarily driven by the aerospace and medical industries due to its ability to fabricate complex geometry parts and provide significant benefits such as weight reduction and customization. However, the nature of AM process results in defects in the final part and understanding the fundamental mechanism is necessary to fabricate parts for critical loading carrying applications. While defects can affect the tensile, toughness, and corrosion properties, they are particularly detrimental to fatigue-critical properties. Hence, it is necessary to establish the process-defect structure-fatigue relationships in AM.

A substantial literature has accumulated on 3D printing with L-PBF that addresses defect structures as a function of the process parameters. Particularly relevant to this work, DebRoy et al. (2018) highlights the variation of properties with processing parameters among other factors and the need for better understanding of the process to minimize defects and tailor microstructure. Along the same lines, Oliveira et al. (2020) also highlights the effect of processing variables such as power, velocity, hatch spacing, and layer thickness on part porosity and discusses the competing interactions between these variables on final part porosity. In the light of community's efforts in establishing

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process-structure-property relationships for AM parts, there is also a substantial literature on the mechanical properties of materials printed via L-PBF, including fatigue. Lewandowski and Seifi (2016) emphasize that it is important to minimize processing defects to elicit the importance of microstructural features that govern fatigue behavior. Furthermore, a two part review by Molaei et al. (2020); Pegues et al. (2020) discusses the powder-processing-structure-property relationships for AM Ti-6Al-4V. Pegues et al. (2020) highlighted the possibility of encountering defects in high density parts and there could be potential variation in the defects and microstructure within in the process window with a variation in processing parameters and powder conditions. The authors also highlight the presence of complex microstructure and defects in the AM material compared to the wrought material that can make fatigue prediction a non-trivial problem. The follow-on review by Molaei et al. (2020) highlights the importance of shape and location of the defect in addition to the defect size. The same authors also reported the use of extreme value statistics to predict the fatigue life based on maximum defect size and reported acceptable agreement with the experimental measurements in 85% of the cases. A recent review by Sanaei and Fatemi (2021) also highlights that research efforts are still underway in fully understanding the effects of defects such as defect distribution and variation on fatigue behavior and being able to predict the fatigue performance. The same authors also discuss the effectiveness of extreme value statistics in predicting the fatigue life. Similarly, Zerbst et al. (2021) opines that research on defect classification and quantification are in their early stages and require further work.

Edwards and Ramulu (2014) performed fatigue testing on L-PBF fabricated Ti-6Al-4V samples and found lower performance compared to the wrought material due to defects in as-fabricated parts such as rough surfaces, residual stress, and porosity. In efforts to parse out the effects of these defects, this section discusses specific example literature related to porosity, as-fabricated surface roughness, post-process optimization treatments including HIP for bulk defects and surface treatments, and applicability of coupon-level testing to final components. Gockel et al. (2019) tested as-fabricated surfaces for fatigue properties and reported an inverse relationship between the maximum surface notch and the fatigue life. In the same work, the authors also highlighted that average roughness, a popularly used measurement for machined surfaces does not strongly correlate with the fatigue life. In recent efforts to accurately correlate surface roughness to fatigue life irrespective of the surface treatment, Lee et al. (2021) have developed a hybrid surface roughness parameter that combines mode, skewness, kurtosis, and maximum valley depth. On the other hand, Eric et al. (2013) performed high cycle fatigue tests on Ti-6Al-4V as-built, polished, and shot-peened specimens to investigate the effect of surface conditions and internal defects on fatigue limit. The polished specimens were found to have highest fatigue limit, which is comparable to wrought data, followed by shot-peened, and as-built specimens. The low fatigue limit value of as-built specimens was attributed to poor surface quality. However, the shot-peened specimens had lower fatigue limit than polished specimens, although shot peening is thought to introduce compressive residual stresses on the outer surface thereby increases fatigue resistance. The reason for this deviation is that in case of shot peened specimens, the crack initiation occurred at subsurface defects and hence reduced the time to failure. Similarly, Lee et al. (2021) also reported an improvement in fatigue strength with laser polishing and stress relief and observed crack initiation at the subsurface defects. To improve the fatigue performance and ductility of additively manufactured Ti-6Al-4V, Kasperovich and Hausmann (2015) performed a two stage optimization procedure which involved reducing internal porosity by optimizing processing parameters, followed by thermomechanical treatments. They found that high temperature HIP post treatment is highly effective compared to simple heat treatments because it results in favorable microstructural changes as well reduces residual porosity thereby resulting in fatigue resistance and ductility comparable to that of wrought alloy. Nevertheless, the number of fatigue tests conducted is limited in this investigation. This means a combination of surface treatment and HIP can potentially address the effect of surface, sub-surface, and residual stress defects on fatigue. Finally, latest work by Leuders et al. (2017) investigated the transferability of mechanical properties of AM lab-scale specimens to application based structures and then compared the properties against parts manufactured through established investment casting processing route. They attributed the superior performance of LPBF-processed Ti-6Al-4V under monotonic loadings to fine microstructure because of high cooling rates. However, they found that the fatigue performance of investment cast is superior to LPBF-processed material in case of both lab-scale specimens and application based structures. They concluded that though post treatments like HIP and shot peening reduce the porosity and residual stresses, the dominant mode of fatigue failure initiation continues to be at single defects. Whereas, in case of investment casting alloy, the damage mechanisms are largely microstructural and hence they exhibit better fatigue performance.

1.1. Process window definition

A schematic of the process window assumed in this work is shown in Fig. 1. There are three boundaries to the process window within which one expects to print to (almost) full density. It is challenging to discuss all the aspects of process window and the reader is referred to a comprehensive study by Gordon et al. (2020). To the bottom right, lack-of-fusion (LoF) is predicted when the melt pools do not overlap sufficiently, and some regions are left unmelted. The pores are typically elongated with sharp corners and contain unmelted particles. Thijs et al. (2010) showed that volumetric energy density (power/velocity.hatch spacing.layer thickness), abbreviated as VED, could be related to component density. Similarly, Read et al. (2015) also studied the influence of multiple process parameters using response surface optimization methodology and VED. They further showed that apart from laser power and scan speed, the interaction between scan speed and hatching speed significantly affects part porosity. On the other hand, Tang et al. (2017) presented a melt pool geometry-based model to predict LoF porosity based on melt pool size measurements, hatch spacing, and layer thickness and demonstrated it on AlSi10Mg. This model was further employed by Islam et al. (2022) to validate the LoF region in process map, which was predicted using a single dimensionless number for nickel superalloy. This closed form criterion results in a sharp boundary

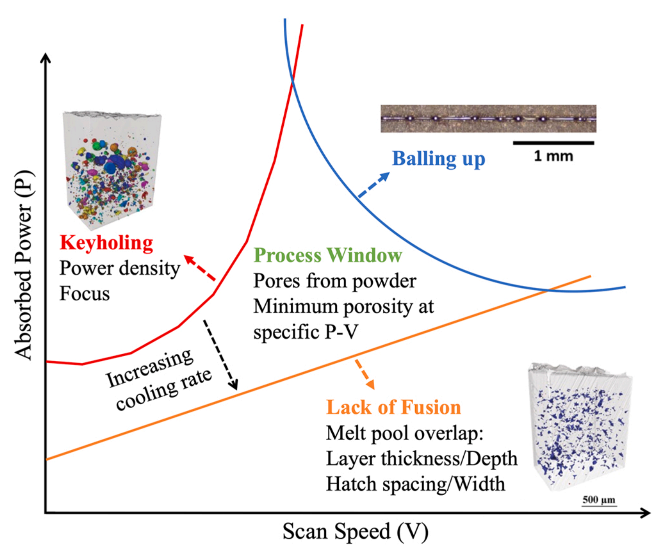

Fig. 1. Schematic process window in power-velocity space, bounded by keyhole porosity (top left), lack-of-fusion (bottom right) and bead-up (top right). Within the process window, porosity is partly inherited from the feedstock powder. Figures from (Cunningham et al., 2017, 2017b) and (Francis, 2017).

in the process space. This means reduced parameters such as linear (power/velocity) or VED omit essential information and require additional modification. For instance, a recent work by Salandari-Rabori et al. (2021) has combined energy density and enthalpy models to develop a reliable VED model to guide process optimization for minimal porosity. Given the simplicity of the geometric model and its dependence on measurable melt pool dimensions and primary processing parameters, the authors in the present work have adopted the model by Tang et al. (2017).

The boundary to the upper left corresponds to the onset of keyhole porosity, i.e., the shedding of pores from the base of deep, narrow vapor cavities formed under conditions of high power density by Zhao et al. (2020). The inset tomograph shows an example of the typically spherical pores found in this condition. King et al. (2014) derived a criterion for keyhole threshold and showed that normalized enthalpy model can adequately estimate the onset of transition from conduction mode melting to keyhole mode melting using material properties and processing parameters. Further, prior work by Francis (2017) investigated the relationship between beam spot size and keyhole porosity and showed that keyholing can be minimized by increasing spot size, resulting in an expanded optimal processing window. On the other hand, in a recent work, Tumkur et al. (2021) demonstrated that using Bessel beams instead of conventional Gaussian-shaped beams reduces keyhole formation tendency across process space, which was attributed to the Bessel beam profile that avoids excessive laser intensity concentration at the center of the melt pool. On the modeling side, to analyze and predict both keyholing and keyhole porosity, Bayat et al. (2019) developed a calibrated and validated numerical model capturing various interconnected physical phenomena along with Fresnel absorption to account for multiple laser reflections.

In addition to porosity, there is increasing evidence that keyhole instability also strongly promotes spatter formation, i.e., the ejection of droplets of liquid metal and displacement of powder. Ly et al. (2017) reported molten spatter from the high temperature front rim of the melt pool and entrained spatter due to the pressure difference between the low-pressure region near the laser beam location and the environment. Gunenthiram et al. (2018) attempted to quantify spatter formation and correlate it to melt pool behavior which is largely governed by process conditions. However, they define both stable and unstable process regimes based on the volumetric energy density values. They found increased volumes of spatter with higher VED and that spatter can largely be suppressed by using lower VED and larger spot sizes. However, VED fails to account for layer thickness and hatch spacing parameters. Hence, it can't be scaled to different powder thickness or materials. Further, Guo et al. (2018) explored the effects of environment pressure on entrained spatter and suggested strategies such as pre-sintering the powder bed and making optimal choice for variables such as powder layer thickness and environment pressure. Overall, keyhole boundary could have significance for both keyhole porosity and spatter formation, although the latter is out of scope of the current work.

The third boundary in the top right direction corresponds to the bead-up instability of the melt pool; this is, however, less well-defined and the underlying physics is not settled. Based on the current knowledge, there could be two possible reasons for balling. To begin with, Li et al. (2012) found that insufficient melting and bonding to the previous layer due to poor wettability can result in irregular tracks. This occurs at low powers and high velocity combinations and thus overlaps with the LoF window. On the other hand, the boundary shown in Fig. 1 is based on the work by Francis (2017), in which the author reported that an optimal selection of melt pool length to width ratio can mitigate balling defect in cases with sufficient melting into the powder layer. This, in turn, was based on the balling criterion developed by Gratzke et al. (1992) for welding processes. In summary, although current literature points to a possible window for the occurrence of balling, this needs further investigation.

Finally, within the process window, the porosity level can be less than 0.1% but the typical gas atomized powder contains internal porosity, part of which carries over to the print (Cunningham et al., 2017, 2017b). In addition to the work done by authors, such process mapping techniques were also implemented by other researchers to identify appropriate processing parameters. For instance, Gong et al. (2014) performed an extensive study to relate defect formation mechanisms to processing parameters in additively manufactured Ti-6Al-4V and divided the process map into four regions based on defect characteristics. Similarly, Kamath et al. (2014) found that the number of experiments required to determine optimal processing parameters can be significantly reduced by combining results from the computational Eagar-Tsai model and single-track experiments. Aoyagi et al. (2019) recently attempted to further reduce the number of experiments required to construct process maps. They classified surface morphology of as-built samples using support vector machine classifier based on the assumption that surface quality is representative of the internal defect content and directly related to processing parameters. Johnson et al. (2019) proposed a computational methodology for predicting a process window for alloys. They developed a finite element based thermal model which accounts for phase-dependent thermophysical properties to accurately estimate the melt pool geometry. It is shown that incorporating phase-dependent properties allows for predicting the changes in melt pool aspect ratios, whereas Eagar-Tsai model has limited capability because of simplified assumptions. The geometry of melt pool is then used to establish thresholds for defect formation mechanisms and to identify ideal process parameters. Seede et al. (2020) also presented an optimization framework that involves mapping out defects structure variation within laser power and velocity processing space by incorporating melt pool geometry prediction from analytical Eagar-Tsai model and single track experiments. To address the uncertainty associated with Eagar-Tsai model predictions, they performed statistical model calibration and utilized the analytical predictions of melt pool dimensions to develop a hatch spacing criterion to avoid generation of LoF porosity. The framework was validated by constructing a processing map for AF9628 alloy. These process mapping development techniques are applicable to various alloy systems and L-PBF machines because these models account for the thermophysical properties of alloys and scatter as result of processing variables.

1.2. Outline of the design of experiment

Fatigue testing remains experimentally intensive as it must typically cover a wide range of alternating stress levels which means that, in general, the print parameters are not varied to any significant extent. One way to rationalize the variations in defect structures at least, is to codify them in a process window referenced to (absorbed) power and scan speed (*aka* velocity) although taking account of other important variables such as hatch spacing, see Fig. 1. It is, accordingly, an advance to investigate the connections between process parameters and defect structures and fatigue behavior and to demonstrate that the process window applies to the mechanical property in the same way as for the defect structures. The preliminary set of results described herein present of order three different parameter sets in each of the main regions, viz. keyhole, LoF, and intra-process window. Defect content was characterized by standard cross-sectioning, polishing, optical imaging, segmentation, and manual counting. For increased throughput in fatigue testing and simplification of machining, 4-point bend fatigue was chosen. All the relevant details are exposed in the subsequent sections. In subsequent publications, a greater density of points (e.g., 300 + samples) will be explored in the process space with the objective of measuring variations in outcomes within the process window, defining how sharp are the edges of the process window. For this report, the hypothesis is that the strong variations found in the defect content are reflected in strong variations in fatigue life.

2. Materials and methods

2.1. Deposition experiments

The Ti-6Al-4V powder used for fabrication of test samples was produced via plasma atomization process by AP&C Powders. The D10-D90 range is 24 – 61 μm with a D50 of 48 μm, and a nominal composition provided by the manufacturer is listed in Table 1. Sample fabrication was performed on an EOS M290 machine at The University of Texas at El Paso (UTEP).

The process conditions were chosen based on the prior work on L-PBF process window development for Ti-6Al-4V by Gordon et al. (2020). Table 2 lists the power and velocity combinations mapping back to the keyhole, process window, and LoF regions. In addition to these, nominal processing conditions recommended by the machine manufacturer for Ti-6Al-4V deposition were also used in this experiment. In summary, four power and velocity combinations were used to fabricate square cross-section bars with dimensions $5 \times 5 \times 73$ mm^3 as shown in Fig. 2. For each power and velocity combination, four samples were fabricated for fatigue testing. Further, the samples were divided into two categories – *moat* and *no moat* samples. A 1 mm thin wall was added 5 mm from the edge to minimize potential temperature build up by adding additional time at the turnaround. Samples with the exterior wall are referred to as *moat* samples and those without the exterior wall are referred to as *no moat* samples in the following discussions. Within each category, four samples were fabricated for each power and velocity combination resulting in a total of thirty-two samples. All these samples utilized a constant hatch spacing of 140 μm, layer thickness of 30 μm, preheat of 180 °C, and a 67° scan rotation between layers. The machine used for fabrication undergoes regular service and maintenance by the machine manufacturer and the operator to ensure process stability. Hence, the authors don't anticipate significant variations in the power and velocity settings during the experiment. However, it is important to note that these quantities can be monitored during the experiment depending on the processing software version installed on the system. Fig. 2 shows the as-fabricated samples on the build plate. It can be observed that deposition was interrupted on some of the samples, and they didn't build completely. Three primary reasons for this failure were swelling in samples with excess energy input, failure at part-support interface, and breaking of supports. These failed samples were not analyzed further.

2.2. Postprocessing

A survey of industrial users, ULI program participants, and literature on Ti-6Al-4V alloys revealed that a range of temperatures from 550 °C to above 800 °C have been recommended or published. Owing to the variety of recipes encountered, the team discussed the needs for the project and came to consensus on using 2 h ± 0.25 at 593 °C (1100°F) followed by air or furnace cooling, with vacuum preferred would eliminate a large fraction of the residual stress while minimizing the risk of oxygen absorption ("hard case"). The furnace temperature tolerance is ± 14 °C (± 25°F). This specification was primarily derived from SAE (AMS2801, 2003).

The team followed the ASTM (2021) standard for fatigue bar preparation that calls for small removals until 125 μm left, then 50 μm per pass for flats with a final 25 μm with longitudinal polishing to $R_a < 0.2$ μm. Overall, 0.5 mm was removed from each face of as-fabricated $5 \times 5 \times 73$ mm^3 square bars and included a 45° chamfers on each edge to eliminate any surface porosity and roughness induced by the contouring. The machining alignment is chosen such that all scratches are parallel to the length of the bar to minimize the tendency to initiate cracks from machining marks. A small number of samples were hand-polished and subjected to 4-point bend fatigue to check for sensitivity to surface condition. A particular company was selected to perform the machining (i.e., low stress grinding) on a regular basis for all the samples fabricated and tested in this program.

Table 2
Processing conditions used in sample fabrication.

Parameter	Power (W)	Velocity (mm/s)	Expected porosity type
A	370	600	Keyhole
B	370	1400	Process window with minimal defects
C	370	3000	Lack-of-fusion (LoF)
D	280	1200	Process window with minimal defects

2.3. Fatigue testing

4-point bend tests were utilized to probe sensitivity of fatigue performance to porosity content and thereby building confidence that a process window can be defined in terms of this critical mechanical property. 4-point bend tests facilitate the use of (i) simple bend bar design, (ii) both as-deposited and machined samples with a simple setup, and (iii) one long bar from which multiple samples can be extracted to test repeatability and variation within one sample. These tests were performed on MTS servo-hydraulic machines at Case Western Reserve University in the Advanced Manufacturing and Mechanical Reliability Center (AMMRC). The requirement of maintaining contact between the fixture and the sample led to the choice of R= 0.1, i.e., wholly tensile stress, and the frequency was fixed at 20 Hz. The length of the sample was chosen such that each sample can be utilized to perform two tests as illustrated in Fig. 3. In addition, finite element simulations were performed to determine that the maximum stress region is located within the 10 mm window, i.e., the tensile stress on the (upper) surface is uniform within 10% in the window.

2.4. Defect characterization using metallography

The broken sections following the fatigue test were polished utilizing the standard sample preparation procedures. Cross-sectional images were captured using a Sensofar S neox surface profiler at a magnification (10X) with a resolution (0.69 μm/pixel). The polishing and imaging steps were repeated several times to obtain cross-sectional images at fixed intervals of ~500 μm. Next, the porosity in the resulting cross-sectional areas was evaluated using the ImageJ software package (Schneider et al., 2012). Pores with an equivalent spherical diameter (ESD) equal to and above 5 μm was considered in the analysis. This lower limit was determined based on the resolution of the profiler. The characterization purpose is to identify the quantity, size, and location of pores within samples. This information combined with extreme value analysis (EVA) will aid in identifying the most critical pore size that can be potentially related to the fatigue behavior.

3. Results

3.1. Fatigue-based process window

The goal of the project is to determine whether the process window

Table 1
Ti-6Al-4V powder composition measured using ASTM (2022).

Element	Al	C	Fe	H	N	O	V	Y	Ti
wt%	6.39	0.02	0.21	0.002	0.01	0.14	3.81	< 0.001	Balance

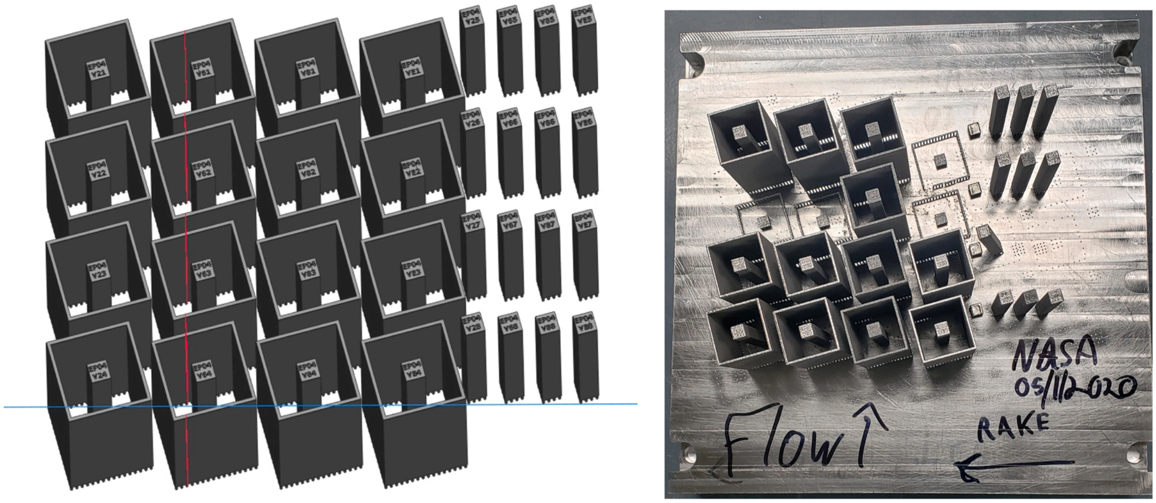

Fig. 2. Schematic of the build layout. (left) Programmed layout and (right) Samples on the build plate after fabrication indicating the gas flow and recoating direction.

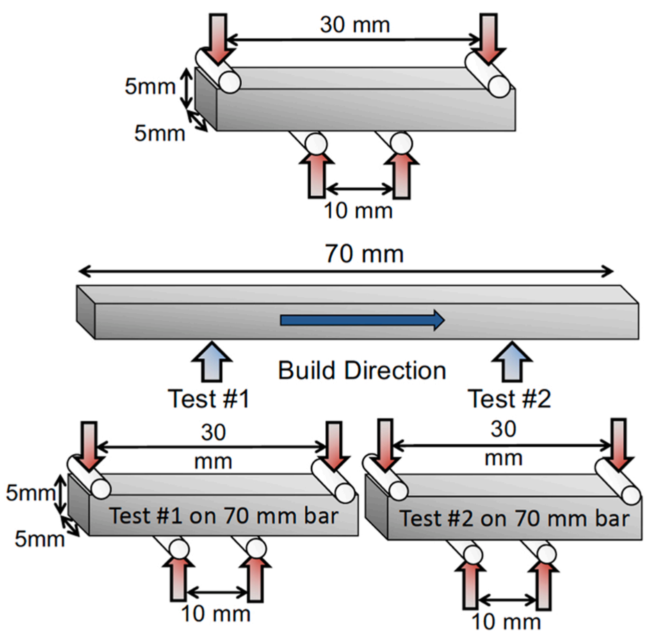

Fig. 3. 4-point bend test specimen geometry and testing setup. A sufficiently large part can be cut into two sections for four-point bend fatigue testing.

outlined by defect content corresponds to the expected variations in fatigue resistance. Testing was conducted in 4-point bending at R= 0.1 on square bars printed standing up, i.e., vertical, as described above. The results obtained to date on samples across P-V space along with typical appearance of fracture initiation sites are summarized in an S-N plot, Fig. 4. The individual points denote the number of cycles to failure at the peak stress noted. A right-pointing arrow denotes a run-out, i.e., the test was stopped at 10^7 cycles. Three different groups of samples are identified by the colored bands to guide the eye, each of which encloses a different print condition. The colored bands are only provided to capture this initial data and do not represent any statistical analysis. This will be included in future reports where 300 + samples have been tested and show very similar results. Each parameter variation has more than one sample, denoted by "repeat" following the sample designation in Table 2. For low-power and high-speed conditions (C), LoF porosity is expected, and this corresponds to the (light yellow) band at the lowest peak stress levels. For high power and low speed conditions (A), keyhole porosity is expected corresponding to one of the bands (light orange) at moderately low stress. The highest stress levels (i.e., 1067 MPa, 1200 MPa) correspond to the (light gray) band for printing conditions inside the process window (B and D) and approach the yield stress reported for as-fabricated and stress-relieved L-PBF Ti-6Al-4V samples in a comprehensive review on mechanical properties of AM materials by. They summarized the yield strength from various reports and reported values ranging between 900 and 1150 MPa with some effects of L-PBF machine and the loading direction compared to the build direction. The average microhardness of these samples exceeded 4.2 GPa while the finite element analysis conducted on these samples confirmed that these peak stresses in the 4PB fatigue tests are only experienced on the outer $5 \times 10\ mm^2$ surface area. This along with the smaller sampling volume compared to axial fatigue testing, will produce higher fatigue run out stresses and better S-N fatigue performance than that reported for axial fatigue in the literature. The S-N behavior of the poor performing samples (i.e., Keyhole and LoF) are in the range of those reviewed in the literature for defected samples, although upcoming axial fatigue tests will also likely reduce these values for the same reasons.

Prior work by Hinderdael et al. (2017) reported peak stresses varying between 800 and 900 MPa for wrought Ti-6Al-4V and 1200 MPa for laser powder fed directed energy deposited Ti-6Al-4V when tested using the four-point bend procedure. The initiation sites included roughness and defects for the wrought samples and defects for the AM samples. Compared to the literature, the AM samples in the current work and the prior work have similar peak stress levels that are approaching yield stress, while the peak stress for the wrought samples is relatively lower. On the other hand, Chern et al. (2020) reported significantly lower stress and lower number of cycles to failure for both electron beam PBF manufactured and wrought Ti-6Al-4V with internal holes. The number of cycles varies between 10,000 and 50,000 depending on the surface conditions. Interestingly, the S-N curves for the process window samples in the current work indicate superior fatigue performance compared to the sand blasted, shot peened, and laser peened L-PBF Ti-6Al-4V samples tested in a recent study by Aguado-Montero et al. (2022). Specifically, sandblasted samples with surface roughness higher than the roughness measured in the current work had multiple initiation points on the surface and failed at 695 MPa and 7226 cycles, which is within the LoF band in the current work. On the other hand, at similar stress levels of 799 MPa, laser and shot peened samples with spherical and LoF pores, respectively failed after 84,338 and 45,992 cycles which qualitatively falls within the keyhole and LoF bands reported in the current work. Furthermore, sample that has undergone shot peening and chemical assisted surface enhancement but containing a spherical pore on the

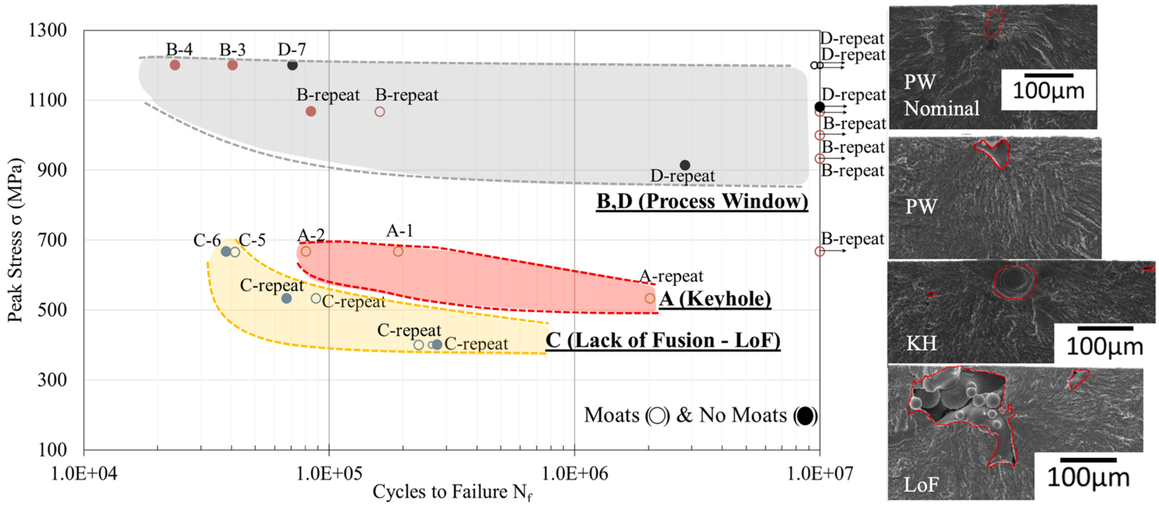

Fig. 4. Summary of 4PB fatigue tests conducted on samples across P-V space along with typical appearance of fracture initiation sites.

order of 40 μm in diameter performed like the sample in the keyhole band. All these prior works utilizing four-point bend tests demonstrate that the results that are reported in the current work follow similar trends reported in the literature. The authors intend to report a more comprehensive comparison in the future publication primarily focused on the fatigue testing part of the broader ULI effort.

All four print conditions are identified in the small insert showing demarcated regions in P-V space. Inset on the right edge of the figure are SEM (scanning electron microscope) micrographs of the area on the fracture surface corresponding to the initiating flaw for selected samples. For LoF conditions, the initiating flaw is large, irregular in shape, and contains unmelted particles. For keyhole conditions, the initiating flaw is also large but circular in cross section as expected for vapor bubbles. For the example from the B condition inside the process window, the initiation does not contain a pore but appears to be of microstructural origin, which raises the intriguing possibility that for a low enough defect content, the fatigue testing may be probing microstructural features that promote crack formation. It is important to note that these bend fatigue tests do not offer any simple approach to determining the number of cycles to initiation. Although detailed analysis of the results will be reported separately, the large variations in effective S-N curve with process conditions points to a strong sensitivity to defect content as originally postulated.

3.2. Defect-based process window

Based on the S-N plot in Fig. 4. Summary of 4PB fatigue tests conducted on samples across P-V space along with typical appearance of fracture initiation sites., select samples were characterized to quantify porosity. The following two scenarios are of interest in this discussion – (i) samples with different peak stress and similar number of cycles to failure (C6 and B3) (ii) samples with similar peak stress and different cycles to failure (C5, A2, and A1; B4, B3, and D7). In the latter category, samples with moats and no moats were separated to ensure any possible changes in porosity content due to variation in part geometry caused by the addition of moat. Each parameter variation has more than one sample and each sample was imaged at multiple locations along the length of the fatigue bar to capture sufficient porosity information. The availability of sufficient data is especially challenging in samples within the process window due to low porosity. Fig. 5 shows a qualitative variation in porosity as a function of processing conditions. The near-spherical pores are typically observed in the keyhole samples and the irregularly shaped pores are observed in the LoF samples, while the samples in the process window have minimal porosity and it is possible that the observed cross-section didn't contain any resolvable pores. Further, Table 3 details corresponding porosity metrics. The purpose of this table is to show the variation in the number of pores. As expected at a constant power of 370 W, an increase in velocity from 600 mm/s to 3000 mm/s first resulted in a decrease in porosity at 1400 mm/s followed by an increase in porosity at higher velocities due to incomplete

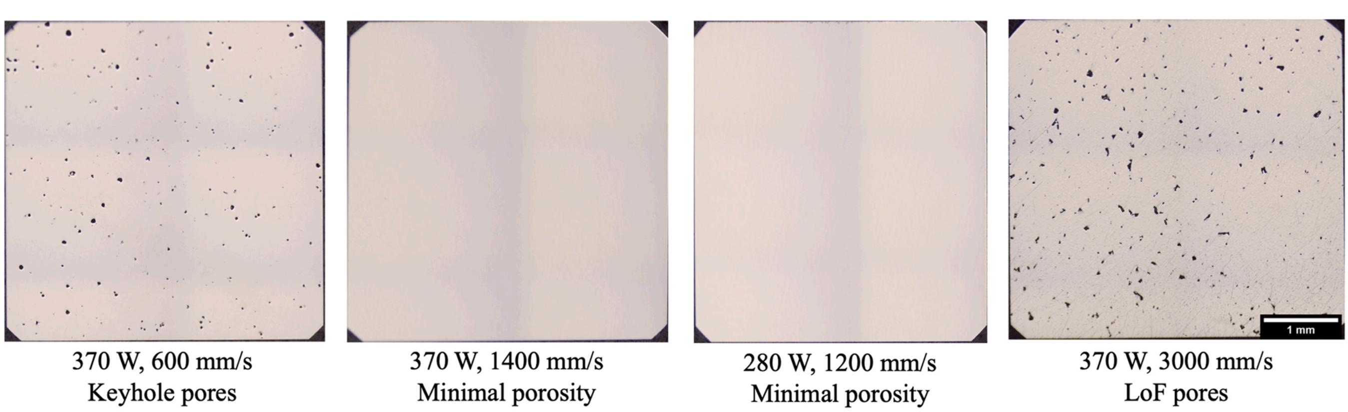

Fig. 5. Representative cross-sectional micrographs (5 ×5 mm^2) captured from samples fabricated using varying power and velocity combinations resulting in different types and levels of porosity. The 45° chamfers are evident on each of the sample corners.

Table 3
Summary of porosity metrics from the select samples characterized in this study considering pores with ESD> 5 μm.

Parameter	Sample Number	Power (W)	Velocity (mm/s)	No. of cross sections	Total No. of pores	No. of pores/ cross-section
A	1	370	600	4	1143	286
	2	370	600	4	1178	295
B	3	370	1400	10	530	53
	4	370	1400	10	485	49
C	5	370	3000	10	10535	1054
	6	370	3000	10	10758	1076
D	7	280	1200	4	42	11

fusion of the powder layer. The sample with nominal processing conditions recommended by the EOS resulted in the lowest number of pores per cross section. This indicates that the expected process regimes used in the design of experiments is repeatable.

Fig. 6 shows the pore size distribution of the samples considering all the pores captured in the segmentation step. There is a shift in the overall distribution with change in processing conditions. A closer look at the upper tail of these distributions shows the difference in extents of the measured maximum pore size with change in processing conditions. Specifically, there is a discernible separation between the upper tail of the samples in the process window region (B3, B4, and D7) containing the lowest number of pores per cross-section and samples from the keyhole and LoF region. This observation also aligns with the fatigue results reported in Fig. 4, where samples in the process window have higher peak stress or higher number of cycles to failure compared to the keyhole and LoF samples. In addition, fractography has shown that the crack initiation sites are microstructural in the process window samples compared to porosity-initiated cracks in the LoF and keyhole conditions.

For the keyhole (A1 and A2) and LoF (C5 and C6) samples, the difference in the upper tail behavior is not as clear from Fig. 6 compared to the process window samples. Hence, extreme value analysis (EVA) was performed on the samples from keyhole and LoF regions to achieve a better understanding of the similarities and differences in the upper tail and relate it to the fatigue behavior in these samples. Beretta (2021) provides a detailed review of adoption and development of statics of extreme for fatigue assessment and its applicability to AM defects. EVA involves identifying the threshold pore size to separate the upper tail

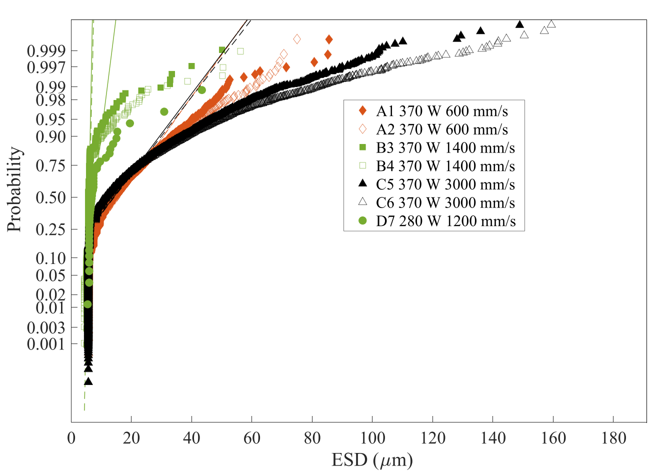

Fig. 6. Normal probability plot of pore sizes from a subset of fabricated samples. A1 & A2 (red) correspond to keyhole conditions, C5 & C6 (black) to LoF, and B3, B4 and D7 (green) to the process window. Although there is a small separation at the median (50%), the upper tails show substantial differences. The deviation from the straight lines indicates that the measurements don't conform to a normal distribution.

from the rest of the pore size distribution and then fit a generalized pareto distribution (GPD) distribution to this data. Prior work, Tang and Pistorius (2019) compared the prediction accuracy of block maxima (BM) and peaks over threshold (POT) methods using defect size distributions obtained via 2D cross-sectional microscopy. They reported accurate predictions using POT sampling strategy which considers all defect sizes above threshold thereby reducing the probability of missing extreme defect sizes. Whereas Romano et al. (2017) applied POT method to defect distribution obtained using non-destructive characterization technique, i.e., X-ray micro-computed tomography. They discuss the influence of defect density assumptions on predictions of the maximum defect size and showed that the application of POT to CT data yields in predictions which align well with the experimental data. Hence, POT analysis was adopted in this work.

Defining the threshold in this method is of high importance in the POT method. In this analysis the threshold was chosen based on mean residual life plot's linear region and stability in GPD shape and scale parameter plots. Table 4 lists the model parameters used for the samples studied in this work and Fig. 7 shows the pore size distribution plots from pores that were determined to belong to the upper tail of the pore size distribution using the threshold. The fitted models were then tested to make sure they result in less than 10% normalized root mean square error (NRMSE) and show good agreement with the empirical data.

The resulting GPD models were then used to generate return value plots. The return level plots corresponding to the fitted GPD models are shown in Fig. 8. In this problem, return level refers to the value of the maximum pore size that can be expected when examining a certain number of pores. The number of pores expected in the maximum stress region on the top surface of the 4-point bend sample was calculated by multiplying the pore density with the estimated maximum stress area (5×10 mm^2) for this sample geometry. Pore density is defined as the number of pores per unit area and calculated using the measurements summarized in Table 3.

Fig. 9 shows a plot of the maximum pore size estimate, expected number of pores in the maximum stress region, and number of cycles to failure from fatigue tests. The maximum pore size is estimated by reading the maximum expected pore size corresponding to the expected number of pores in return level plots. From further examining Fig. 9, it can be observed that maximum expected pore size from EVA could clearly distinguish between the LoF and keyhole parameters. In addition, the maximum expected pore size from EVA and the crack initiation pore size measured from the fracture surface are in good agreement. This demonstrated the applicability of EVA to predict the range of crack initiating pore sizes in the stressed region. However, within the keyhole conditions, for similar maximum expected pore size, the number of cycles to failure vary significantly. This will require further investigation to understand other factors such as the presence of clustered pores, surface vs. sub-surface initiation, etc., that could possibly affect the fatigue behavior in addition to the initiating pore size.

4. Discussion

4.1. Transferability approach applied to a high-strength AlMgZr alloy

This discussion provides a summary of the main issues encountered when transferring parameter development approach across materials and indicate that the approach is indeed transferable with the need to

Table 4
Threshold, scale, and shape parameters for GPD fit.

Sample Designation	Threshold (μm)	Scale	Shape
A1(keyhole)	11	13.98	-0.1204
A2 (keyhole)	11	14.56	-0.1265
C5 (LoF)	29	15.34	-0.0115
C6 (LoF)	26	17.12	0.0367

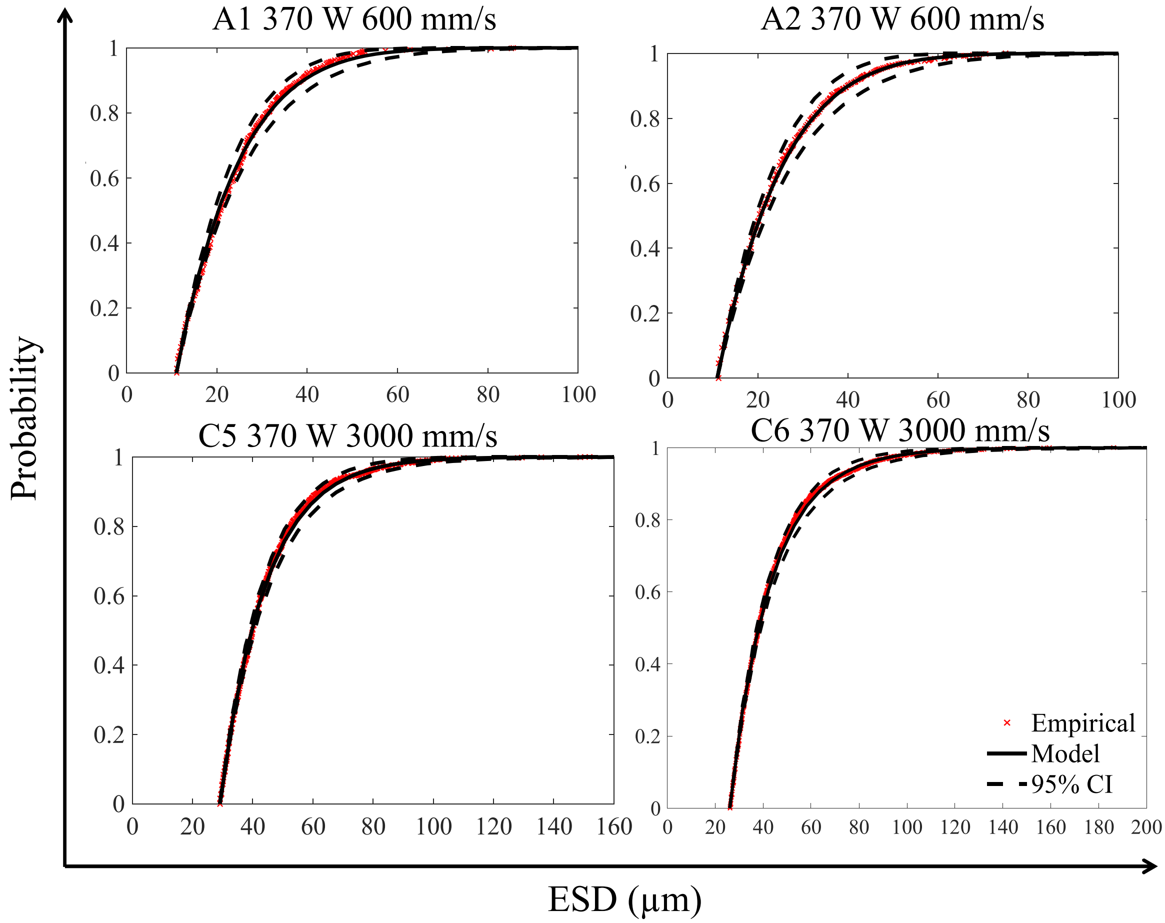

Fig. 7. Cumulative distribution function (CDF) plots generated from GPD models fitted to the upper tail of the size distributions in Fig. 6 for the keyhole condition (A1 & A2) versus LoF condition (C5 & C6).

calibrate absorptivity via single track beads. Materials Resources, LLC (MRL), is a small business service bureau whose role in the ULI project has been to transfer the process mapping technique that was developed for Ti-6Al-4V on an EOS M290 machines to a novel high strength aluminum alloy for AM (AlMgZr) printed on a 3DSystems Prox200 L-PBF machine. In this project, MRL acted as a manufacturer responding to the customer specification to demonstrate technology transition from CMU to a service bureau by completing builds and delivering porosity and melt pool analytics for process parameter sets designed by the rest of the team.

Firstly, the process map boundaries were calculated according to prior work on Ti-6Al-4V by Gordon et al. (2020), and process parameter sets were selected based on these boundaries, see the workflow in Fig. 10. In planning for the first build with the AlMgZr alloy, material constants obtained from literature on AlSi10Mg were used to estimate the melt pool dimensions that are required to predict process map boundaries. The resulting porosity measured by MRL reflected process map boundaries that were significantly shifted from the predictions. By completing an additional build of single tracks at various power and velocity combinations, MRL was able to provide melt pool width data to recalculate the material constants for this specific combination of material and printer. This resulted in more accurate, calibrated process map predictions that were validated by a subsequent build and porosity analysis at MRL. Throughout this technology transfer effort, the critical takeaway was the necessity of a calibration step when applying the process mapping models to a new material and machine. The authors intend to share a full description of this work along with the results in the future publications.

4.2. Effect of powder batch

Prior work by Narra et al. (2020) on electron beam powder bed fusion demonstrated that with adjustments in process parameters such as power, velocity, hatch spacing, and spot size, hydride-dehydride (HDH) powders that are non-spherical and irregular in shape can be used to fabricate parts with similar quality to the ones fabricated using spherical powder. Furthermore, Wu et al. (2021) reported similar findings when using HDH powders in the L-PBF process. Based on these studies the authors expect that for a new powder batch with different morphology and size distribution, the process window shifts, and adjustments must be made to achieve high part density. This means consistent results can be obtained if the powder size distribution, composition, and quality are reasonably the same. In ongoing work as part of the broader ULI effort, the effect of change in powder batch, location of printing, and across different machines is being studied. The authors intend to share these findings in future publications.

4.3. Thermal history

The intrinsic fatigue strength of the L-PBF processed Ti-6Al-4V (i.e., with no porosity or minimal porosity such as the process window samples) depends on its microstructure, which is a function of its thermal history, and hence it is of great importance to model the process-microstructure relationship in an L-PBF part as a first step toward predicting the intrinsic fatigue life. The more data-driven approach is to note that, although the large variation in fatigue life is clearly determined by the defect content, part of the remaining variation may be explainable from the dependence of microstructure on thermal history.

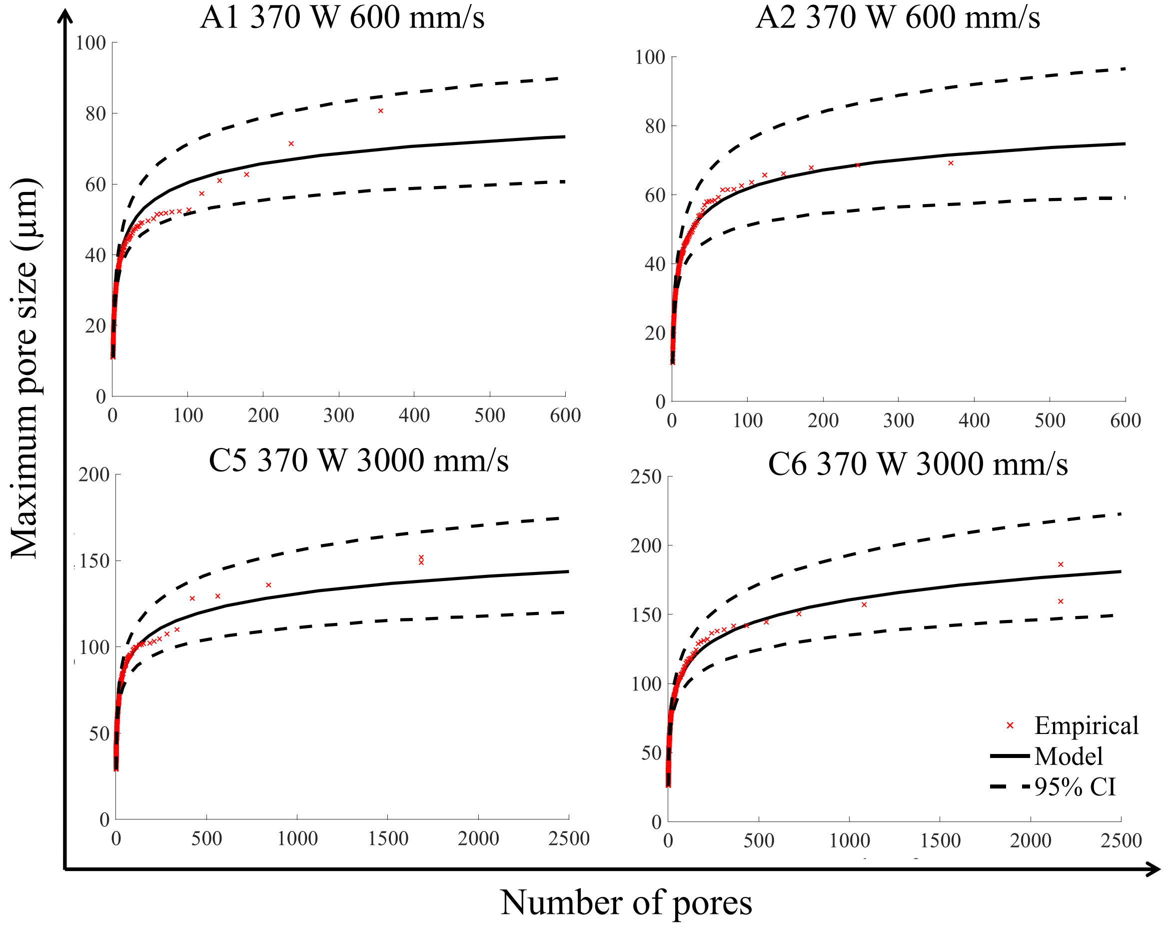

Fig. 8. Return level plots to estimate the maximum expected pore size in a larger region of interest for the keyhole condition (A1 & A2) versus LoF (C5 & C6).

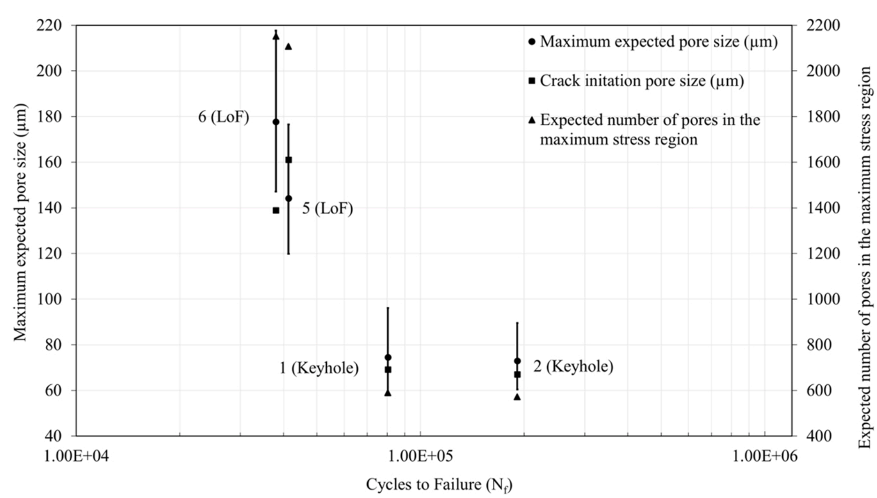

Fig. 9. Maximum expected pore size, crack initiation pore size measured from fracture surface, and number of pores in the maximum stress region plotted against the reported cycles to failure from fatigue tests for samples fabricated using the keyhole and LoF conditions. The peak stress is 667 MPa.

In general terms, cooling rate is inversely related with energy input per unit length, which means that conditions close to keyholing have low cooling rates whereas those close to LoF have higher cooling rates, see Fig. 1.

One approach to achieve this is to compute the detailed temperature history using scanwise thermal process simulation following the exact laser scanning path and utilize it to predict the volume fractions of the different phases with a mean-field metallurgical phase transformation model. In the phase transformation model, the solid-state diffusional kinetics is modelled using the Johnson-Mehl-Avrami-Kolmogorov (JMAK) equation, while the kinetics of martensitic transformation (diffusionless) are modeled using the Koistinen and Marburger (KM) equation. Baykasoğlu et al. (2020) used this approach and has successfully employed to predict the detailed thermal history and microstructure in a small Ti-6Al-4V part consisting of 20 single tracks processed by a laser powder directed energy deposition (DED) process. However,

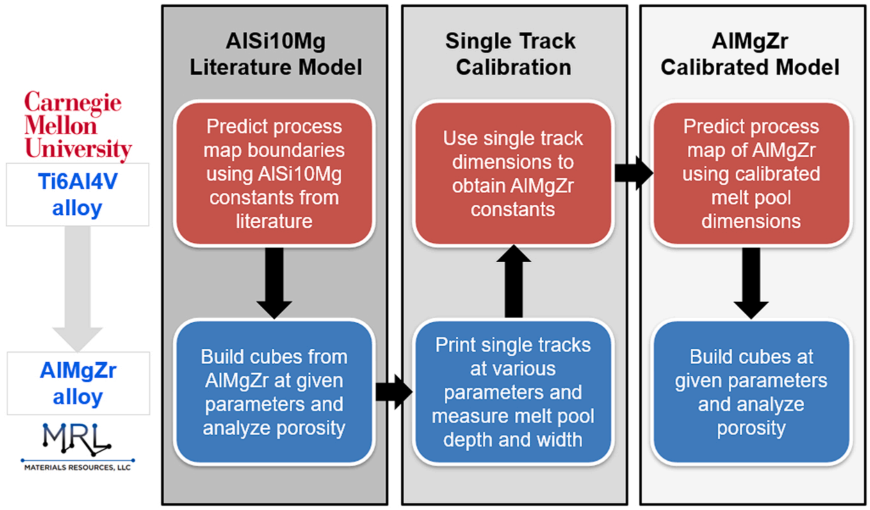

Fig. 10. Collaboration workflow between Materials Resources, LLC (MRL) and CMU for technology transfer to a high strength aluminum (from Ti-6Al-4V) and 3D Systems ProX200 machine (from EOS M290).

scanwise process simulation following the laser scanning path is computationally intractable for L-PBF part-scale simulation involving millions to billions of scan lines. In a review on modeling challenges and opportunities, Francois et al. (2017) suggest that the simulations should capture sufficient physics to practically model kilometers of laser scans during the layer-by-layer fabrication. It may be feasible to employ a global-local analysis, where a part-scale layerwise simulation can be performed first over the entire part, and several locations of interest can be selected for high-resolution microscale scanwise simulations. To make the part-scale simulation efficient, a layerwise approach will be adopted where the entire numerical layer is activated and heated simultaneously. Then a detailed thermal history can be computed at selected locations of interest using the scanwise simulation of a millimeter-scale region. The thermal boundary conditions on the microscale simulation domain will be subjected to the temperature time histories obtained from the part-scale simulation at the locations of interest.

With the advancement in Graphical Processing Unit (GPU) technology, both the part-scale and microscale process simulation models can take advantage of GPUs to significantly speed up the simulations on relatively inexpensive workstations. The layerwise thermal process simulation of a complex geometry using the GPU-accelerated process simulation approach demonstrated by Dugast et al. (2020) is presented here, see Fig. 11. The size of the baseplate used in the simulation is $50 \times 50 \times 12$ mm^3 and the size of the part is $50 \times 50 \times 50$ mm^3. The mesh resolution is 50 μm along the build direction and 250 μm in x and y directions. Hence there are 200×200 elements in the x-y plane and 1000 layers of elements in the build direction. To reduce computational cost, an adaptive meshing technique is implemented, where elements with 50 μm layer thickness are used in the top eight layers, while for the layers below, the mesh is coarsened automatically to a height of 250 μm in each layer as shown in right panel of Fig. 11. The default EOS M290 DMLS processing parameters for Ti-6Al-4V and temperature-dependent properties are used in the simulation. A simulation time of 3.3 h for this complex part is achieved using the simulation approach running on a workstation with four NVIDIA Titan V GPU cards. This simulation time is comparable with the actual build time of 3 h. This example demonstrates that it is very much feasible to utilize GPU-accelerated process simulation to simulate the detailed thermal history and phase transformation in a complex L-PBF Ti-6Al-4V part.

4.4. *Guidelines for qualification of L-PBF Printers*

The destination of this work is a physics-based approach to qualification of L-PBF printers. The basic motivation is both evidential, i.e., minimal porosity when printing inside a process window with corresponding optimal fatigue resistance over the same range, and physical, i.e., the strong evidence for keyhole instability ergo porosity at high power plus low speed and LoF on the other side. This realization leads directly to guidelines for qualification that focus on determining the position of the process window, along with checking that its borders are in reasonable locations relative to prior work on similar materials.

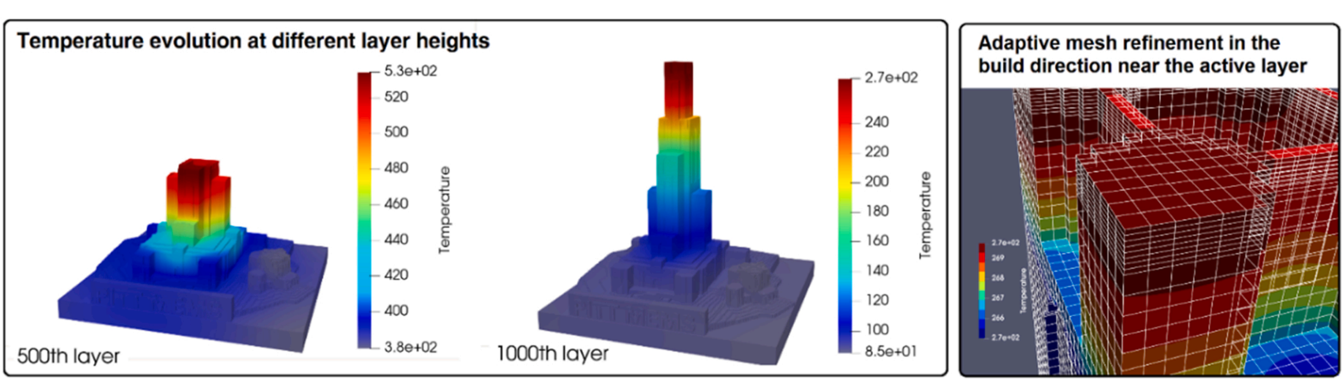

Fig. 11. Thermal process simulation of the Cathedral of Learning of the University of Pittsburgh using the GPU-accelerated process modeling method: (left) Temperature profile at a height of 500 layers and 1000 layers and (right) Adaptive mesh refinement at top of the model.

Maintenance of the process window should be based on tracking changes in position of the borders over time. The most likely reasons for changes include degradation of the laser optics over time (i.e., delivery of less power) and changes in the feedstock powder that cause changes in, e.g., packing density. Measurement of the variations in porosity as a function of parameters can be accomplished via printing test cubes which should be no smaller than a centimeter on a side. As Fig. 10 above shows, an essential component of the qualification process is to check the dimensions of single-track beads, which is a way of measuring the variation in absorptivity. The latter is expected to vary strongly as a function of laser power density, i.e., the greater the extent of keyholing (i.e., depth of the vapor cavity), the greater the fraction of laser power absorbed, which is especially true in materials with low intrinsic absorptivity such as copper and aluminum.

Scime and Beuth (2019) reported variability in melt pool geometry (e.g., width and depth) across process space. Therefore, it is important to quantify the variance in addition to the average values because LoF can occur wherever *any* insufficiency in overlap occurs (which may explain the occurrence of so-called "rogue flaws"). The significance of power density also motivates the importance of tracking laser power and spot size over time. Lastly, it is important to note that the "recipe" supplied by the printer manufacturer is likely to be within the process window, even if it was determined empirically, but there is no reason why the user should not adopt a recipe that is more nearly in the center of the process window. Moreover, maximizing print rate generally requires use of higher scan speed and concomitant higher power, which provides a further motivation for determination of the process window. With respect to variations in local conditions during an extended build, there are many examples in the literature of heat build-up, which was one of the motivations for the thermal modeling discussed in the section above on thermal modeling. In other words, one must, in practical printing, expect local temperature variations that run the risk of effectively taking the conditions outside the process window. The best approach to this challenge is to measure the process window location as a function of pre-heat.

5. Conclusions and future work

Gaps still exist in establishing the process-critical defect/microstructure-fatigue relationships, which is the primary focus of the present work. A thorough understanding of the contributions of defect and microstructure to fatigue can be accomplished by the framework in the present work utilizing variation in processing conditions based on a fundamental understanding of the defect formation mechanisms. This work provides a compact overview of the physics-based process qualification approach to establish process-defect structure-fatigue relationships in 3D printing of metals. Our conclusions from the current work are as follows.

- 4-point bend fatigue testing was demonstrated to be effective as a screening technique for distinguishing between different defect content levels, to be followed with axial fatigue testing of selected sample conditions. Lack-of-fusion (LoF) conditions resulted in the worst fatigue performance, with keyhole porosity giving slightly better performance. The best performance was recorded for printing in the center of the process window and conditions towards LoF but still within the process window showed poorer performance.
- For the power and scan speed levels available in the L-PBF printer used, the variation in defect content from keyholing to (near) full density to LoF conditions were in accord with prior experience. This indicates that porosity variations are reproducible and consistent with the known physics of the process.
- Guidelines for L-PBF qualification have been developed that focus on a 2-stage protocol. The first stage involves measuring the melt pool size as a function of power and speed while maintaining other parameters such as spot size (function of focus settings on the equipment), layer thickness, and preheat constant. This information combined with hatch spacing is used to predict the boundaries of the process window. The second stage involves printing test coupons with different parameter combinations designed to cover the process window; these coupons must of course be sectioned, polished, imaged and analyzed for porosity content. Although average pore area fraction (equivalent stereologically to volume fraction) is sufficient to establish the process window, extreme value analysis is recommended for connection to properties such as fatigue. In addition, local temperature variations could be addressed by incorporating preheat as an additional variable during the development of the process windows. The need for this depends on the part geometry under consideration.
- The process window approach was successfully transferred to a different location, printer type and material (aluminum alloy, as opposed to Ti-6Al-4V). The initially assumed absorptivity was, however, found to be incorrect and emphasized the need to use single melt track experiments to calibrate the models used to predict the process window, most particularly for the LoF boundary which is directly tied to melt pool overlap.
- A combination of layerwise and scanwise simulation utilizing GPU-accelerated process simulation approach will play a key role in understanding the microstructural changes caused by thermal history variations during the fabrication. This could possibly explain fatigue behavior observed in samples fabricated using optimal processing conditions resulting in low porosity.

The results presented here point to many potential extensions. The efficacy of 4-point bend fatigue as a screening technique would allow quantification of the effect on the process window of many other parameters relating to both printing and materials. The use of extreme value analysis, while known in the fatigue community, could broaden out from being primarily a modeling tool to being based on experimental data. For instance, Przybyla and McDowell (2012) utilized extreme value statistics on non-local fatigue indicator parameters calculated using the polycrystalline plasticity models to understand the competing effects of surface and sub-surface defects in Ti-6Al-4V. On the other hand, Urbano et al. (2015) applied extreme value statistics on experimental data from the cross sections and fracture surfaces to develop empirical correlations between the extreme inclusions in Nitinol to the fatigue limit. Pursing the latter clearly depends on the development of more efficient methods for assessing defect (porosity) content, given that both computed tomography and metallography on cross-sections are time-consuming and thus expensive. Community-based pooling of characterization and mechanical test data represents an opportunity for the application of Findability, Accessibility, Interoperability, and Reusability (FAIR) principles reported by Frazier et al. (2021). Although the preliminary results shown here suggest comparable performance for the same type of printer in different locations, such round robin testing needs to continue to increase confidence in the process window approach to qualification. Although computational engineering is represented here only via simulation of thermal history, there are clear opportunities for using integrated computational materials engineering (ICME) to predict many aspects of defect generation in 3D printing such as keyhole instability, spatter, melt pool size & shape, and micro-mechanical response.

CRediT authorship contribution statement

Sneha P. Narra: Conceptualization, Methodology, Writing – original draft, Visualization, Supervision, Resources. **Anthony D. Rollett:** Conceptualization, Methodology, Writing – original draft, Resources, Project administration, Funding acquisition. **Austin Ngo:** Writing – original draft, Formal analysis, Investigation, Validation. **David Scannapieco:** Writing – original draft, Formal analysis, Investigation, Validation. **Mahya Shahabi:** Software, Formal analysis. **Tharun Reddy:**

Writing – original draft, Validation. **Joseph Pauza:** Investigation. **Hunter Taylor:** Investigation. **Christian Gobert:** Investigation. **Evan Diewald:** Investigation. **Florian X. Dugast:** Writing – original draft, Formal analysis. **Albert To:** Writing – original draft, Supervision. **Ryan Wicker:** Writing – review & editing, Conceptualization, Methodology, Supervision, Resources. **Jack Beuth:** Writing – review & editing, Conceptualization, Methodology, Supervision, Resources. **John J. Lewandowski:** Conceptualization, Methodology, Writing – original draft, Supervision, Resources.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

The raw/processed data required to reproduce these findings cannot be shared at this time as the data is currently being used in another study.

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